



Title of the Final Year Undergraduate Project

An undergraduate project report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

**Degree of the Bachelor of the Science of Engineering
Honours**

by

R. Rathnaweera	-	RU/E/2005/004
A. Abekon	-	RU/E/2005/005
B. Bandara	-	RU/E/2005/006
C. Kankalu	-	RU/E/2005/007

.....
Prof. A.B.C. Dee
(Supervisor)

.....
Prof. A.B.C. Dee
(Co-Supervisor)

Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decisionmaking. Current State-of-the-Art deep learning models often rely on posthoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multiclass classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be backprojected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

Acknowledgments

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Preface

This thesis is submitted.....

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Acronyms

CXR	-	Chest X-Ray
CT	-	Computed Tomography
CAD	-	Computer-Aided Diagnosis
CNN	-	Convolutional Neural Network
ViT	-	Vision Transformer
XAI	-	Explainable Artificial Intelligence
GGO	-	Ground-Glass Opacity
COPD	-	Chronic Obstructive Pulmonary Disease
SotA	-	State of the Art

Chapter 1

Introduction

Developing a unified pipeline for multi-class abnormality classification and grounded localization addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is accountable and its claims are supported by clinical evidence.

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality [1]. Research shows that early detection of pulmonary diseases significantly improves patient outcomes. For this, radiological imaging tools such as Chest X-Rays (CXR) and Computed Tomography (CT), plays an important role. However, the increasing volume of such imaging data increases radiologists' workload, leading to delays and potential errors in diagnosis. This has motivated the development of automated Computer-Aided Diagnosis (CAD) systems, which use computational algorithms and Artificial Intelligence (AI) to assist in tasks such as abnormality detection, classification, and segmentation [2].

Our research mainly focuses on the development of a CAD system using 3 Dimensional (3D) CT data. While CXR is commonly used as the first imaging test due to its low cost and low radiation exposure, studies show that their sensitivity to abnormalities is limited [3]. Comparatively, CT scans produce high resolution cross-sectional slices that helps detect smaller, early-stage, and subtle lung pathologies that are frequently missed by standard CXR.

However, CT analysis is complex and time-consuming, requiring radiologists to review hundreds of slices per scan, which are often inconsistent due to artifacts, anatomical variations and different scanners/protocols. Furthermore, different anatomical structures can share identical grayscale values. Telling them apart requires expert clinical judgment which is often subjective. High sensitivity can lead to detection of clinically insignificant abnormalities, while visual fatigue can lead to missed findings [4].

To address these challenges, recent research and clinical deployments of AI-based CAD systems are fundamentally changing how radiologists approach CT analysis. AI models such as 3D Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have shown to reduce interpretation time, and improve lesion detection sensitivity compared to traditional manual methods. Many of these high perform-

ing models are Deep Learning based, which are criticized for their black box nature, where their internal reasoning is difficult to interpret. This lack of interpretability is a significant barrier for clinical adoption, as radiologists require evidence to support a model’s diagnostic claims.

This has motivated the development of explainable AI (XAI) methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) that helps to visualize exactly which parts of an image a model focuses on when making a specific prediction. Such XAI methods are highly valuable for building clinical trust in deep learning models. However, they often lack the fine-grained, pixel-level precision required to identify tiny lesions, sometimes highlighting incorrect or confounding features [?].

Recent advances in sparse representation models such as Sparse Autoencoders and Dictionary Learning have shown to provide inherent interpretability in medical imaging. This stems from their ability to reconstruct complex, high-dimensional data as a linear combination of ”atoms” or features. This allows the use of efficient algorithms to easily map the output sparse domain back to the original input feature space, providing a linear relationship between the model’s decision and anatomical structures. Our work builds on these advances by applying dictionary learning to 3D Chest CT data for multi-class abnormality classification and grounded localization.

1.2 Problem Statement

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model’s decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this study is to address the limitations of existing AI-based approaches for abnormality classification and localization in chest CTs, by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning based CAD system using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.

- Develop a unified evaluation framework for assessing the performance and interpretability of the proposed approach.
- Benchmark model performance against SotA classification and segmentation models on the same dataset.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 2 thoracic abnormalities: Pulmonary Nodules, and Ground-Glass Opacities, based on expert radiologist recommendation and dataset availability.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to interpretable methods, and will not explore approaches that provide better performance for less explainability.

Chapter 2

Literature Review

2.1 Clinical Background

Chronic respiratory diseases account for more than 4 million deaths globally per year [1]. Research shows that early detection of pulmonary diseases, particularly lung cancer and Chronic Obstructive Pulmonary Disease (COPD) dramatically lowers mortality rates. Two landmark studies, the National Lung Screening Trial (NLST) and the Netherlands-Leuvens Longkanker Screenings Onderzoek (NELSON) trial, have shown that low-dose CT (LDCT) screening reduces lung cancer mortality by 20% to 33% compared to CXR [2, 3].

2.1.1 GGO and Pulmonary Nodules

LDCT is commonly used to detect abnormalities such as pulmonary nodules and GGO in the lungs.

What are pulmonary nodules and ground-glass opacities (GGOs)? Why are they clinically important? Why is early detection essential? Why are these abnormalities difficult to detect?

2.1.2 Computed Tomography (CT)

What role does CT play compared to other imaging modalities? What challenges exist in interpreting 3D chest CT scans?

2.2 Computer-Aided Diagnosis (CAD) Systems

What is Computer-Aided Diagnosis? What tasks does CAD perform? * classification
* detection * localization * segmentation

2.2.1 Evolution of CAD Systems

How have CAD systems evolved? Why has deep learning become dominant?

2.3 Deep Learning Approaches

General idea on CNNs, ViTs, and other deep learning architectures.

How well do they perform? Which datasets are commonly used? What evaluation metrics are reported?

2.3.1 nnU-Net

2.3.2 BioMedParse

BiomedParse is a foundation model for segmentation, detection, and recognition that extends prompt driven parsing to medical imaging tasks [?]. Rather than relying on a fixed label set during inference, the model accepts a natural language prompt describing the target anatomy or pathology and produces a pixel level mask for the queried concept. This makes BiomedParse especially useful in settings where the structures of interest are heterogeneous, difficult to enumerate in advance, or described differently across institutions and reports.

In the context of chest CT, BiomedParse is attractive because many clinically relevant findings such as lung nodules, lung opacity, consolidation, and atelectasis can be queried directly with disease specific text prompts. The model combines image encoding with language understanding and uses a transformer based decoder to align the prompt with the visual features extracted from the scan [?]. The output is not only a segmentation map but also a form of interpretable localization, since the generated mask indicates where the model believes the prompted abnormality is present. Therefore, BiomedParse serves both as a SotA reference model and as a practical contrast for discussing performance, localization quality, and interpretability in 3D chest CT analysis.

2.3.3 TotalSegmentator

Accurate anatomical segmentation is an important step in CT-based computer-aided diagnosis (CAD). Earlier deep learning methods were mainly developed to segment only a few organs. To solve this problem, Wasserthal *et al.* introduced **TotalSegmentator**, a deep learning framework that can automatically segment 104 anatomical structures from a single CT scan [?]. The framework is built on the *nnU-Net* architecture, which automatically adjusts the training process to different datasets. This helps the model achieve reliable segmentation on different clinical CT images.

The model was trained using more than 1,200 routine clinical CT scans collected from different hospitals and patient groups. It achieved a mean Dice Similarity Coefficient (DSC) of about 0.94, showing high segmentation accuracy. Because of its good performance, TotalSegmentator is widely used as a preprocessing tool in medical image analysis, including computer-aided diagnosis, surgical planning, and radiotherapy planning [?].

Although TotalSegmentator provides accurate anatomical segmentation, it is not designed to classify pulmonary abnormalities such as lung nodules or ground-glass opacities. It also does not explain how its predictions are made. Therefore, additional methods are needed for disease classification and interpretable clinical decision-making.

The proposed research aims to overcome these limitations by developing an interpretable Label Consistent K-SVD (LC-KSVD) framework for pulmonary abnormality classification and grounded localization. While TotalSegmentator provides

useful anatomical information, the proposed framework focuses on improving both classification performance and the interpretability of the results.

2.3.4 Merlin

2.3.5 MedSAM

2.3.6 CT-CLIP

CT images contain valuable information for diagnosing pulmonary diseases, while radiology reports provide detailed clinical descriptions. To learn information from both sources, Hamamci *et al.* proposed **CT-CLIP**, a vision-language foundation model for chest CT images [?]. Unlike traditional supervised models that require large amounts of labeled data, CT-CLIP learns the relationship between CT scans and their corresponding radiology reports using contrastive learning. This allows the model to learn meaningful image and text representations without task-specific annotations.

CT-CLIP consists of a 3D image encoder and a text encoder. The image encoder extracts features from CT volumes, while the text encoder processes the corresponding radiology reports. During training, the model learns to match each CT scan with its correct report in a shared feature space. The model was trained on thousands of chest CT scans paired with radiology reports, allowing it to learn both anatomical and pathological information. It has shown strong performance in pulmonary disease classification, image retrieval, and zero-shot learning tasks [?].

Figure 2.1: Overview of the CT-CLIP framework showing contrastive learning between CT images and paired radiology reports (adapted from [?]).

Although CT-CLIP achieves good performance, it mainly learns high-level feature representations instead of directly localizing pulmonary abnormalities. It also does not provide inherent explanations for its predictions, making the decision process difficult to understand.

The proposed research aims to overcome these limitations by developing an interpretable Label Consistent K-SVD (LC-KSVD) framework for pulmonary abnormality classification and grounded localization. While CT-CLIP learns rich image and text representations, the proposed framework focuses on providing transparent and interpretable diagnostic results.

2.4 Explainable AI (XAI) in Medical Imaging

Artificial intelligence (AI) has become an important tool in medical imaging by supporting tasks such as disease detection, classification, and image analysis. Many deep learning models achieve high accuracy, but their decision-making process is often difficult to understand. These models are commonly referred to as "black-box" models because they do not clearly explain how a prediction is made [? ?]. In healthcare, this lack of transparency can reduce clinicians' confidence in AI-assisted diagnosis.

Interpretability is important because medical decisions directly affect patient care. Clinicians need to understand why a model predicts a particular disease before using

its results in clinical practice. Explainable AI (XAI) provides methods that help users understand how AI models make decisions. This improves trust, supports clinical decision-making, and helps identify possible model errors or biases [?].

The importance of XAI has also increased due to ethical and regulatory requirements. Healthcare AI systems are expected to be transparent, reliable, and accountable before they are used in real clinical environments. Explainable models help meet these requirements by providing evidence that supports their predictions instead of producing results without explanation [? ?].

Explainability is especially important for chest CT diagnosis because pulmonary abnormalities such as lung nodules and ground-glass opacities can be difficult to identify. Clinicians need to know which regions of the CT scan influenced the model's prediction. This helps verify the diagnosis and increases confidence in AI-assisted clinical decisions. Therefore, improving interpretability has become an important research direction in computer-aided diagnosis systems [?].

2.4.1 SHAP

SHapley Additive exPlanations (SHAP) is a widely used explainable artificial intelligence (XAI) method for interpreting machine learning models. It is based on Shapley values from cooperative game theory, where each feature is assigned an importance value according to its contribution to the final prediction. This provides a consistent and theoretically sound approach for explaining both traditional machine learning and deep learning models [?].

One of the main advantages of SHAP is that it provides both global and local explanations. Global explanations identify the features that have the greatest overall influence on model predictions across the entire dataset. Local explanations show how each feature contributes to the prediction of an individual sample. SHAP values can be positive or negative, indicating whether a feature increases or decreases the predicted outcome. These explanations help users understand why a model makes a particular decision and improve confidence in AI-based systems [? ?].

In medical imaging, SHAP has been widely used to improve the transparency and interpretability of artificial intelligence models. It helps researchers and clinicians identify the imaging features and clinical variables that have the greatest influence on model predictions. This makes it easier to determine whether the model is learning meaningful medical information rather than irrelevant patterns. As a result, SHAP supports the development of more reliable and trustworthy computer-aided diagnosis systems [? ?].

For pulmonary disease analysis using CT images, SHAP can explain the factors that influence a model's predictions by measuring the contribution of each input feature. These explanations help clinicians understand AI-assisted decisions, verify whether the model focuses on clinically relevant information, and identify possible prediction errors. Therefore, SHAP improves the interpretability, reliability, and clinical acceptance of deep learning models used for pulmonary disease diagnosis [? ?].

2.4.2 LIME

Local Interpretable Model-agnostic Explanations (LIME) is a popular XAI method used to explain the predictions of machine learning models. It is model-agnostic,

which means it can be used with any type of model, no matter how it works internally. LIME explains one prediction at a time by building a simple model that approximates how the original model behaves near that prediction [?].

For image data, LIME first divides the image into small regions, called superpixels. In medical images, these regions can also be meaningful patches of the scan. LIME then creates many versions of the image by hiding or changing some of these regions and checks how the model’s prediction changes each time. A simple linear model is then fitted to these results, and the regions with the highest importance are shown as the explanation [? ?].

In medical imaging, this helps researchers and doctors see which parts of an image the model used to make its decision. This makes it easier to check if the model is looking at real medical features instead of irrelevant patterns [? ?]. For pulmonary disease detection using CT scans, LIME can point out the regions linked to abnormalities, helping clinicians confirm that the model is making sensible decisions and catch possible errors [?].

However, LIME also has some drawbacks. Its explanations can change between runs because of the randomness in how it perturbs the input, and the results depend on settings like the number of samples used [?]. It is also slow to run on large 3D CT scans, since each perturbed sample needs a separate prediction from the model. Because of this, gradient-based methods like Grad-CAM are often preferred for 3D medical imaging tasks [?].

2.4.3 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a popular XAI method used to explain the predictions of convolutional neural networks (CNNs). It is not model-agnostic like LIME. It only works with CNN models, because it uses the gradients from the last convolutional layer to find out which parts of the image the model looked at [?].

Grad-CAM works by looking at the feature maps from the last convolutional layer. It calculates the gradient of the predicted class score with respect to these feature maps. These gradients are averaged to get an importance weight for each feature map. The feature maps are then combined using these weights and passed through a ReLU function, so only the parts with a positive effect on the prediction are kept. This gives a coarse heatmap that shows the regions of the image that mattered most for the prediction [?].

In medical imaging, Grad-CAM is used a lot because it is fast and can be applied to any CNN without changing the model. It helps check if a deep learning model is looking at real abnormalities, like tumours or lesions, instead of random background areas [? ?]. For pulmonary disease detection using CT images, Grad-CAM can highlight areas linked to nodules or ground-glass opacities (GGOs), helping doctors confirm the model’s decision is based on meaningful areas [?].

However, Grad-CAM has some limitations. Since it uses the last convolutional layer, its heatmap has lower resolution than the input image and needs to be upsampled. This can make the explanation blurry and less accurate for small abnormalities [?]. It can also struggle when there are multiple objects of the same class in an image, since it may highlight them all together instead of separately [?].

2.4.4 Grad-CAM++

Grad-CAM++ is an improved version of Grad-CAM. It was made to fix some of Grad-CAM’s problems, especially when an image has more than one object of the same class, or when the object is small [?].

The main difference is how the importance weight for each feature map is calculated. Grad-CAM uses a simple average of the gradients. Grad-CAM++ uses a weighted combination of higher-order derivatives instead. This gives more importance to the pixels that contribute the most to the prediction, instead of treating all pixels equally [?]. Because of this, Grad-CAM++ can locate objects better when there are multiple instances of the same class, and it can highlight the full object more accurately than Grad-CAM.

In medical imaging, this is useful when there is more than one abnormality of the same type in a scan, like multiple lung nodules in one CT slice. Grad-CAM++ can separate and highlight each nodule clearly, giving a fuller explanation than Grad-CAM, which may only focus on the biggest one [? ?]. For pulmonary CT analysis, this means Grad-CAM++ can help doctors see all the important abnormal areas in a scan, not just the most obvious one. This matters because missing a small nodule could lead to a missed diagnosis.

However, Grad-CAM++ also has drawbacks. It costs more to compute than Grad-CAM, since it needs to calculate higher-order gradients [?]. It also has the same resolution problem as Grad-CAM, since it also relies on the last convolutional layer and needs upsampling to match the input image size [?].

2.4.5 Other CAM Methods

Besides Grad-CAM and Grad-CAM++, several other CAM-based methods have been proposed to address specific weaknesses in the original approach. Three common ones are HiResCAM, Score-CAM, and XGrad-CAM.

HiResCAM was introduced to fix a problem with how Grad-CAM averages gradients. This averaging step can sometimes make Grad-CAM highlight regions that the model did not actually use for its prediction. HiResCAM solves this by multiplying the gradients with the feature maps directly, instead of averaging the gradients first. This produces an explanation that is more faithful to what the model actually used to make its decision [?]. HiResCAM has been applied to 3D medical imaging tasks, such as detecting abnormalities across CT slices, where faithful localisation is especially important [?].

Score-CAM takes a different approach. Instead of using gradients at all, it measures the importance of each feature map by masking the input image with that feature map and checking how much the model’s confidence score changes. Feature maps that cause a bigger increase in confidence are given a higher weight [?]. Because it does not use gradients, Score-CAM avoids some of the noise and instability that gradient-based methods can have. However, it is much slower, since it needs a separate forward pass through the model for every feature map [?].

XGrad-CAM was designed to make Grad-CAM more mathematically consistent. The authors introduced two properties, called sensitivity and conservation, that a good explanation method should satisfy. Grad-CAM does not fully satisfy these properties, so XGrad-CAM changes how the importance weights are calculated to better meet them [?]. In practice, this gives XGrad-CAM better localisation than

Grad-CAM while still being fast to compute, since it does not need the extra higher-order gradients that Grad-CAM++ requires [?].

Each of these methods offers a different trade-off. HiResCAM improves faithfulness with almost no extra cost, Score-CAM improves stability at the cost of speed, and XGrad-CAM improves theoretical grounding while staying efficient. For 3D medical imaging tasks such as pulmonary CT analysis, this means the choice of method should depend on whether faithfulness, stability, or speed matters most for the task at hand [? ? ?].

2.4.6 Limitations of XAI Methods

XAI methods like LIME and Grad-CAM have made deep learning models easier to understand. But they still have some important limitations.

One issue is that many of these methods only give an approximate explanation, not an exact one. LIME builds a simple model around a prediction to explain it. But this simple model may not always match the real model well, especially for complex, non-linear networks [? ?].

Another problem is instability. Running the same method on the same input more than once can give slightly different results. This makes it harder for doctors to trust the explanation if it keeps changing [? ?].

Gradient-based methods like Grad-CAM have their own issues too. They rely on the last convolutional layer, so the heatmap has low resolution and needs to be upsampled. This can make it harder to localise small abnormalities precisely [? ?].

Most XAI methods also only show where the model looked, not why it made that decision. They do not explain the deeper reasoning behind a prediction [? ?]. This is a real problem in medical imaging, since doctors often need to understand the reasoning, not just the location.

Finally, there is no single standard way to measure how good an explanation is. Different XAI methods are often evaluated with different metrics, which makes it hard to compare them fairly [? ?].

Because of these limitations, XAI methods should be used carefully. Their explanations need to be checked before being trusted in real clinical decisions.

2.5 Sparse Representation

Why is sparsity useful? How is sparse representation learned?

2.5.1 Sparse Coding

What is sparse coding? What is overcomplete representation? Why is sparse representation robust?

Two types: Greedy Algorithms and Convex Relaxation

2.5.2 Greedy Algorithms

Matching Pursuit (MP)

Orthogonal Matching Pursuit (OMP)

2.5.3 Convex Relaxation Algorithms

Least Angle Regression (LARS)

Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

2.6 Dictionary Learning

What is a dictionary atom? How is reconstruction performed? Why are atoms interpretable? What optimization algorithms exist?

2.6.1 K-Means Singular Value Decomposition (K-SVD)

2.6.2 Method of Optimized Directions (MOD)

2.6.3 Online Dictionary Learning (ODL)

2.7 Discriminative Dictionary Learning

Traditional dictionary learning methods mainly focus on minimizing the reconstruction error between the input data and its sparse representation [?]. However, good reconstruction does not necessarily produce features that are useful for classification. Samples from different classes may activate similar atoms if the learned dictionary only captures common structures in the data.

Discriminative dictionary learning addresses this limitation by incorporating class information during dictionary optimization. The objective is not only to represent the input samples accurately but also to learn sparse codes that improve the separation between different classes. By learning class aware atoms, discriminative dictionary learning can provide both classification capability and a more interpretable representation. Several approaches have been proposed, including Label Consistent K-SVD (LC-KSVD) and Fisher Discriminative Dictionary Learning (FDDL), which extend the K-SVD framework by introducing supervised learning constraints.

2.7.1 Label Consistent K-SVD (LC-KSVD)

Label Consistent K-SVD (LC-KSVD), introduced by Jiang et al. [?], extends the classic K-SVD framework by embedding label consistency into the dictionary update process. The core idea is to encourage sparse codes from samples of the same class to exhibit similar activation patterns, while those from different classes diverge. This is achieved without sacrificing the algorithm's efficient iterative structure involving sparse coding and dictionary refinement.

The method comes in two main variants. LC-KSVD1 incorporates a label consistency term that aligns the sparse codes with an ideal discriminative structure through a linear transformation:

$$\min_{D, A, A_c} \|X - DA\|_F^2 + \alpha \|Q - A_c A\|_F^2 \quad \text{s.t.} \quad \|a_i\|_0 \leq T \quad \forall i. \quad (2.1)$$

Here, Q is a target matrix encoding class specific indicator patterns, and A_c learns to map codes toward this structure. LC-KSVD2 extends LC-KSVD1 by adding a classifier learning term. The dictionary, sparse codes, label consistency transformation, and classifier are jointly optimized:

$$\min_{D, A, A_c, W} \|X - DA\|_F^2 + \alpha \|Q - A_c A\|_F^2 + \beta \|H - WA\|_F^2 \quad \text{s.t.} \quad \|a_i\|_0 \leq T, \quad (2.2)$$

where H contains the label information. The additional classification term allows the learned sparse representation to be directly optimized for prediction. Due to its ability to learn class specific atoms, LC-KSVD has been applied in image classification tasks where interpretability of learned features is important [?].

2.7.2 Fisher Discriminative Dictionary Learning (FDDL)

Fisher Discriminative Dictionary Learning (FDDL), proposed by Yang et al. [?], is another supervised dictionary learning approach that directly enforces Fisher like criteria on both the dictionary atoms and the sparse coefficients. It learns a structured dictionary $D = [D_1, D_2, \dots, D_c]$ with class specific sub dictionaries D_i while promoting sparse codes A that exhibit low within class scatter and high between class scatter.

The core objective combines a discriminative fidelity term $r(\cdot)$ (ensuring good reconstruction within class and poor reconstruction across classes) with a Fisher discrimination penalty on coefficients:

$$\min_{D, X} \sum_i r(A_i, D, X_i) + \lambda_1 \|X\|_1 + \lambda_2 \left(\text{tr}(S_W(X)) - \text{tr}(S_B(X)) + \eta \|X\|_F^2 \right), \quad (2.3)$$

where $r(\cdot)$ encourages each class to be well represented by its own sub dictionary while poorly represented by others. S_W and S_B denote the within class and between class scatter matrices of the sparse codes. The elastic term controlled by η is added to guarantee that the overall Fisher discrimination function remains convex, which greatly improves optimization stability [?].

Compared with LC-KSVD, which introduces label consistency through a learned transformation and classifier, FDDL focuses on improving the statistical distribution of sparse codes. This makes it suitable for classification tasks where the separation between classes is important.

2.7.3 Frozen Dictionary Learning for Outlier Separation

While most discriminative methods update the entire dictionary during training, certain scenarios benefit from preserving previously acquired knowledge. Frozen dictionary learning, as introduced in the Outlier Learning via Augmented Frozen (OLAF) Dictionaries framework [?], addresses this by fixing a subset of atoms learned from baseline (normal) data while allowing new atoms to adapt to anomalous samples.

The composite dictionary takes the partitioned form $D = [D_f \mid D_u]$, where D_f remains frozen after initial training on normal data, and only the unfrozen portion D_u

is updated on data containing deviations. Both K-SVD and Alternating Minimization algorithms can be modified straightforwardly to respect this constraint by skipping updates on frozen columns during the dictionary refinement stage [?].

This approach is particularly useful when preserving existing knowledge is important. In medical imaging, a dictionary learned from normal anatomy can be kept fixed, while additional atoms are learned from abnormal cases. This separation helps maintain normal structural information and improves the interpretability of disease related atoms. However, the effectiveness of this method depends on the quality of the initial frozen dictionary, since an incomplete initial representation may limit the ability of the model to adapt to new patterns.

2.7.4 Dictionary Learning in Medical Imaging

Dictionary learning has been applied to many areas of medical image analysis over the past two decades. The main idea is to learn a set of atoms directly from the data, so that each image or patch can be represented as a sparse combination of these atoms. This approach was first popularised by the K-SVD algorithm, which learns an overcomplete dictionary by alternating between sparse coding and dictionary update steps [?]. Later, Jiang et al. extended this idea to classification tasks by proposing Label Consistent K-SVD (LC-KSVD), which adds a label consistency term so that the learned dictionary is more discriminative for classification [?].

In medical imaging, dictionary learning has been used for a wide range of tasks. Ravishanker and Bresler used dictionary learning to reconstruct MRI images from undersampled k-space data, allowing shorter scan times without losing image quality [?]. Tong et al. applied discriminative dictionary learning to segment the hippocampus from brain MRI scans, showing that sparse coding can capture useful anatomical patterns for segmentation [?]. In CT imaging, Ben-Cohen et al. combined a fully convolutional network with sparsity-based dictionary learning to detect liver lesions, showing that dictionary learning can also be used alongside deep learning models rather than as a replacement for them [?]. A broader survey by Zhao et al. reviewed the use of dictionary learning for CT image de-noising and histopathological image classification, and applied low-rank shared dictionary learning to glaucoma diagnosis on fundus images [?].

These studies show that dictionary learning methods, and LC-KSVD in particular, remain a relevant classical approach in medical imaging. Their sparse representations offer a natural way to interpret which parts of an image contributed to a decision. This makes dictionary learning a suitable classical baseline for comparison against deep learning-based interpretability methods.

2.7.5 Image Reconstruction

2.7.6 Classification and Segmentation

2.8 Evaluation Metrics

The evaluation of medical image analysis systems requires multiple metrics because different measures capture different aspects of model behaviour. In chest CT analysis, classification metrics describe the ability of the model to identify abnormal findings,

segmentation metrics measure spatial agreement with reference annotations, and interpretability metrics evaluate whether the learned representations provide meaningful explanations.

2.8.1 Classification

For a binary classification problem, let TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. Accuracy measures the overall proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (2.4)$$

Although accuracy provides an overall performance estimate, it can become misleading when the dataset contains class imbalance. Therefore, additional metrics based on positive class prediction are considered.

Precision indicates the reliability of positive predictions by measuring the fraction of detected abnormalities that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (2.5)$$

Recall, also referred to as sensitivity, measures the ability of the model to recover existing abnormal cases:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (2.6)$$

The F1-score combines precision and recall into a single measure and provides a balanced evaluation when both false positive and false negative errors are important:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (2.7)$$

The Area Under the Receiver Operating Characteristic Curve (AUROC) evaluates the ability of a classifier to separate positive and negative samples over all possible decision thresholds. It is calculated from the relationship between the true positive rate (TPR) and false positive rate (FPR):

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR}. \quad (2.8)$$

A higher AUROC indicates that the model can better distinguish between abnormal and normal cases without depending on a particular classification threshold.

Average Precision (AP) summarizes the precision-recall relationship and is particularly useful for medical datasets where abnormal cases may represent a smaller proportion of the data:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \quad (2.9)$$

where P_n and R_n represent precision and recall values at different classification thresholds. Higher average precision indicates better detection of positive cases under class imbalance.

The reliability of predicted probabilities is measured using Expected Calibration Error (ECE). The metric compares predicted confidence with the actual accuracy within confidence intervals:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (2.10)$$

Lower ECE values indicate that the predicted confidence scores better reflect the actual likelihood of correctness.

2.8.2 Localization and Segmentation

For segmentation evaluation, let P represent the predicted region and G represent the ground-truth annotation. The Dice coefficient measures the overlap between the two regions:

$$\text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \quad (2.11)$$

Intersection over Union (IoU), also known as the Jaccard index, measures the ratio between the intersection and union of the predicted and reference regions:

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \quad (2.12)$$

Dice and IoU provide complementary information about segmentation quality, where higher values indicate better spatial agreement between predicted and reference regions.

2.8.3 Interpretability of Sparse Dictionary Learning

In sparse dictionary learning, interpretability is obtained from the learned dictionary atoms and their corresponding sparse representations. Therefore, evaluation focuses on whether the dictionary provides compact, meaningful, and discriminative representations of CT image patterns.

The reconstruction error measures how effectively the learned dictionary represents the input data. Given an input patch x , dictionary D , and sparse coefficient vector α , the reconstruction error is defined as:

$$\text{Reconstruction Error} = \|x - D\alpha\|_2^2. \quad (2.13)$$

A lower reconstruction error indicates that the dictionary captures important structures present in the image data.

The sparsity of the representation describes how many atoms contribute to the reconstruction. It is measured using the number of non-zero coefficients:

$$\text{Sparsity} = \|\alpha\|_0. \quad (2.14)$$

Lower values indicate that only a small number of atoms are required to explain a sample, making the representation easier to interpret.

Dictionary coherence evaluates the similarity between different atoms in the learned dictionary. The mutual coherence is defined as:

$$\mu(D) = \max_{i \neq j} \frac{|d_i^T d_j|}{\|d_i\|_2 \|d_j\|_2}. \quad (2.15)$$

A lower coherence value indicates that the dictionary contains less redundant atoms and that individual atoms capture more distinct image characteristics.

Atom utilization measures how frequently each dictionary atom contributes to the representation of the dataset:

$$U_k = \sum_{i=1}^N \mathbb{I}(\alpha_{k,i} \neq 0), \quad (2.16)$$

where U_k represents the activation frequency of atom k . This metric helps identify whether the learned dictionary is effectively utilized or dominated by a limited number of atoms.

For supervised dictionary learning approaches such as LC-KSVD, class-specific atom activation provides additional interpretability. Disease-related atoms are expected to activate more frequently for samples belonging to their associated class compared with unrelated classes. This indicates that the learned atoms capture discriminative patterns rather than only general image structures.

Finally, visual inspection of reconstructed dictionary atoms is used as a qualitative interpretability assessment. By examining representative atoms, their relationship with anatomical structures or pathological appearances can be analysed, providing an intuitive understanding of the features learned by the model.

2.9 Research Gaps

Chapter 3

Methodology

Chapter 4

Conclusion

Appendix A

Essential English Grammar - A Handout

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

A.1 Simple Present Tense

A.2 Perfect Tense

A.3 Past Past Perfect Tense

A.4 Passive Voice

A.5 Technical Writing Fundamentals

A.6 Proof Reading

Appendix B

Graphical Interpretation of Data

The graphical interpretation given below shows the relationship between x and y is acceptable to the purpose of the project

B.1 Data Analysis

B.1.1 Statistical Analysis

B.1.2 Empirical Analysis

B.2 The X vs Y vs Z

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